

ASTANA IT UNIVERSITY
School of Software Engineering

DEVELOPMENT OF A WEB PLATFORM FOR REAL-TIME AUTHORIZED SEARCH OF ACADEMIC SUPERVISORS

Second Pre-Defence Report

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1. Introduction

1.1 Project Overview

This document constitutes the Second Pre-Defence Report for the capstone project titled "Development of a Web Platform for Real-Time Authorized Search of Academic Supervisors," developed by a three-person team at Astana IT University (AITU) under the School of Software Engineering.

The platform — internally referred to as SupervisorMatch — addresses a documented inefficiency in Kazakhstan's higher-education system: the absence of a centralized, digital mechanism through which students can discover, evaluate, and formally request academic supervision. At present, supervisor selection at AITU and similar institutions relies on informal email outreach, word-of-mouth referrals, and paper-based departmental lists, all of which are opaque, time-consuming, and prone to duplication of requests.

1.2 Institutional Context and Stakeholders

SupervisorMatch is designed as a B2B SaaS product targeting university administrations across Kazakhstan, with AITU serving as the initial pilot institution. Three distinct user roles are supported:

Role	Responsibilities	Key Pain Point Addressed
Student	Search supervisors, view profiles, submit and track supervision requests	Lack of visibility into supervisor availability and research interests
Supervisor (Faculty)	Manage research profiles, set availability/capacity, review and decide on incoming requests	Unstructured, duplicate inbound requests with no tracking tool
Administrator	System oversight, user management, audit reports and analytics	Manual, opaque process with no aggregate data on matching outcomes

This stakeholder definition directly addresses feedback received at the First Pre-Defence, which noted that the institutional context and stakeholder roles were insufficiently articulated.

1.3 Response to First Pre-Defence Feedback

The committee provided the following observations after the first pre-defence:

- The problem is relevant in academic administration; however, the specific institutional context and stakeholders are not clearly defined.
- A detailed requirements specification distinguishing between functional and non-functional requirements is needed.
- Similar platforms used in universities or research institutions should be reviewed.

This report responds to each item in turn. Section 1.2 above defines the institutional context and stakeholders in detail. Section 4 provides a comprehensive requirements specification. Section 3 presents a structured analysis of comparable platforms.

2. Literature Review

2.1 The Supervisor–Student Relationship in Academic Settings

The academic supervisor–student relationship is widely recognised as one of the most consequential factors in a student’s research development and overall academic success. Lindgreen et al. [1] conducted one of the early systematic investigations into how students find and choose supervisors in a marketing research context, identifying three central challenges: lack of accessible information about supervisor expertise, absence of a structured matching process, and the predominantly informal, word-of-mouth nature of supervisor discovery. Their findings established a conceptual foundation for the argument that centralised, information-rich platforms can meaningfully improve matching quality.

More recent empirical work by Yu et al. [2] examined the status and influence of supervisor–student relationships across Chinese universities, finding that the quality of the supervisory relationship directly correlates with student academic output, satisfaction, and timely graduation. Their analysis highlights that poor initial matching—where students are assigned to supervisors with misaligned research interests or insufficient capacity—is a leading cause of supervisory relationship breakdown. These findings directly motivate the design goals of SupervisorMatch: enabling interest-based filtering and enforcing real-time capacity constraints at the point of request submission.

2.2 Supervisor Influence on Student Academic Development

Wu et al. [3] conducted a qualitative investigation into how supervisors affect students’ academic gains and research ability, finding that supervisors who are accessible, responsive, and aligned with students’ research interests generate significantly better academic outcomes than those who are assigned by administrative default. Their study identifies “availability uncertainty”—students not knowing whether a supervisor has capacity to take on new students—as a specific structural barrier that creates anxiety and delays project initiation. This finding provides direct empirical support for the real-time availability display feature central to the SupervisorMatch platform.

Zhong et al. [4] extended this line of inquiry by examining how servant-style supervisory leadership influences postgraduate employability through the mediating mechanisms of student self-efficacy and academic engagement. Their structural equation model demonstrated that supervisory quality has cascading effects well beyond the immediate research project, affecting graduates’ confidence and professional readiness. This evidence reinforces the business case for improving the quality of initial supervisor–student matching: a platform that increases the probability of a good fit at the outset produces downstream benefits that justify institutional investment in digital matching infrastructure.

2.3 The Role of Supervisors in the Research Process

Everitt [5] investigated the appropriate degree of supervisor involvement in students’ literature search and review writing, concluding that supervisors who clearly communicate their engagement style and expectations from the outset produce students with stronger independent research skills and higher-quality outputs. A key implication for platform design is that supervisor profiles should expose not just research interests and availability, but also qualitative information about supervisory approach—a feature identified as a post-MVP enhancement in this project’s roadmap (Section 6).

2.4 Gap Identified in the Literature

Taken together, the literature consistently demonstrates that: (a) the quality of supervisor–student matching is a significant determinant of academic outcomes [1][2][3][4]; (b) supervisor accessibility and availability information are critical inputs to good matching decisions [3]; (c) supervisory communication and style transparency improve student independence and output quality [5]. However, the literature does not document a purpose-built digital platform that addresses all three of these dimensions simultaneously within the context of Central Asian higher education institutions. This gap constitutes the primary justification for the SupervisorMatch project.

3. Analysis of Existing Systems

3.1 Comparative Review

As part of the literature review and requirements analysis phase, the team conducted a structured comparison of four existing platforms that address adjacent problems in academic supervision or research-profile matching. The goal was to identify feature gaps that SupervisorMatch can close within the specific context of Kazakhstani universities.

Platform	Institution / Context	Key Features	Limitations for our context
FindASupervisor (UK)	Multiple UK universities	Keyword search, supervisor profiles with research areas, direct enquiry form	No real-time availability; no request tracking; not integrated with university SSO
ARAP (Academic Research Access Portal)	Australian universities	Research profile registry, publication links, supervisor capacity indicators	No student-side request workflow; read-only for students; no admin oversight dashboard
SupervisorFinder (NZ)	University of Auckland	Department-filtered search, email contact trigger on profile click	No role-based access; supervisor availability not displayed; no request status history
LinkedIn + email (informal)	Widespread globally	Rich profile data, direct messaging	Not purpose-built; no capacity/availability data; no institutional governance; privacy concerns

The comparative analysis reveals a consistent pattern: existing tools provide search and profile-viewing capabilities but lack a governed, end-to-end request workflow with real-time availability, role-based access control, and administrative oversight — features that are central to the SupervisorMatch value proposition.

3.2 Key Differentiators of SupervisorMatch

- Real-time capacity display — supervisors set a maximum student load; availability updates dynamically as requests are accepted.
- End-to-end request workflow — students submit structured requests; supervisors accept or reject; the system records and notifies both parties.
- Role-based access control — privacy of student contact details is enforced until a request is formally accepted.
- Administrative dashboard — department heads and administrators gain aggregate visibility over supervisor utilisation and matching outcomes.
- JWT-based authentication — all users authenticate via a secure token mechanism; future integration with university SSO (Single Sign-On) is planned.

4. Requirements Specification

The following specification was derived through a combination of stakeholder interviews (conducted with three AITU students and two faculty members), review of comparable platforms (Section 3), and iterative refinement during the design sprint in March 2026.

4.1 Functional Requirements

4.1.1 Authentication and Authorization

- FR-AUTH-01: The system shall allow users to register and log in using a valid university email address and password.
- FR-AUTH-02: The system shall issue JWT tokens upon successful authentication and validate them on every protected API request.
- FR-AUTH-03: The system shall enforce role-based access control, distinguishing Student, Supervisor, and Administrator roles.
- FR-AUTH-04: The system shall prevent students from viewing supervisor private contact details until a supervision request has been formally accepted.

4.1.2 Supervisor Search and Discovery

- FR-SRCH-01: Students shall be able to search for supervisors by name, research area, academic department, or keyword.
- FR-SRCH-02: Students shall be able to filter search results by department, research area, and availability status (available / unavailable).
- FR-SRCH-03: Search results shall display at minimum: supervisor name, title, department, current student load (X/N slots), and research area tags.
- FR-SRCH-04: The system shall use PostgreSQL full-text search indexing to return results with sub-second latency under normal load.

4.1.3 Supervisor Profile Management

- FR-PROF-01: Supervisors shall be able to create and edit a research profile containing: bio, research interests, available project topics, prerequisites for applicants, and links to publications.
- FR-PROF-02: Supervisors shall be able to set and update their maximum student capacity and toggle their availability status.
- FR-PROF-03: Supervisor profiles shall be publicly visible to authenticated students.

4.1.4 Supervision Request Workflow

- FR-REQ-01: Authenticated students shall be able to submit a supervision request to any available supervisor, including a brief motivational message.
- FR-REQ-02: The system shall reject a request submission if the target supervisor's capacity is full, returning an informative error to the student.
- FR-REQ-03: Supervisors shall be able to view incoming requests with student name, degree programme, group, and message.
- FR-REQ-04: Supervisors shall be able to accept or reject each request. Accepting shall decrement available capacity by one; rejecting shall leave capacity unchanged.
- FR-REQ-05: Upon acceptance, the student's contact email shall be revealed to the supervisor.
- FR-REQ-06: Both student and supervisor shall receive an in-system notification when a request status changes.

4.1.5 Student Profile and Application History

- FR-STU-01: Students shall maintain a profile with name, degree programme, academic group, and an optional short bio.
- FR-STU-02: Students shall be able to view a history of all submitted requests and their current statuses (Pending / Accepted / Rejected).

4.1.6 Administrator Dashboard

- FR-ADM-01: Administrators shall access a dashboard showing aggregate statistics: total active requests, acceptance rates per department, supervisors approaching capacity.
- FR-ADM-02: Administrators shall be able to manage user accounts (activate, deactivate, change roles).
- FR-ADM-03: The system shall maintain an audit log of all request state transitions, accessible to administrators.

4.1.7 Recommendation System

- FR-REC-01: Upon viewing a supervisor profile, students shall be shown up to three recommended supervisors with similar research areas (based on tag overlap).

4.2 Non-Functional Requirements

ID	Category	Requirement	Acceptance Criterion
NFR-PERF-01	Performance	Search results must return within 1 second under standard load.	95th percentile response time < 1 s in load tests with 100 concurrent users.
NFR-PERF-02	Concurrency	The system must support at least 200 concurrent authenticated users.	No errors or timeouts during a 200-user load test of 10 minutes.
NFR-SEC-01	Security	All API endpoints must be protected by JWT authentication.	Unauthenticated requests to protected routes return HTTP 401.
NFR-SEC-02	Privacy	Student contact data must not be exposed until request acceptance.	API response for unaccepted requests omits email field entirely.
NFR-SEC-03	Data Integrity	All request state changes must be atomic and logged.	No orphaned requests found after simulated concurrent accept/reject operations.
NFR-AVAIL-01	Availability	The platform must maintain 99% uptime during the academic semester.	Downtime < 7.2 hours per semester; monitored via uptime service.
NFR-USAB-01	Usability	A new student user must complete a supervisor search and request submission within 3 minutes without documentation.	Usability test with 5 participants achieves task completion in under 3 minutes.
NFR-MAINT-01	Maintainability	All backend modules must have unit test coverage of at least 70%.	Coverage report generated by Jest shows >= 70% line coverage.

NFR-SCAL-01	Scalability	The system architecture must support horizontal scaling of the backend.	NestJS backend deployable as multiple instances behind a load balancer without state issues.
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5. Methodology

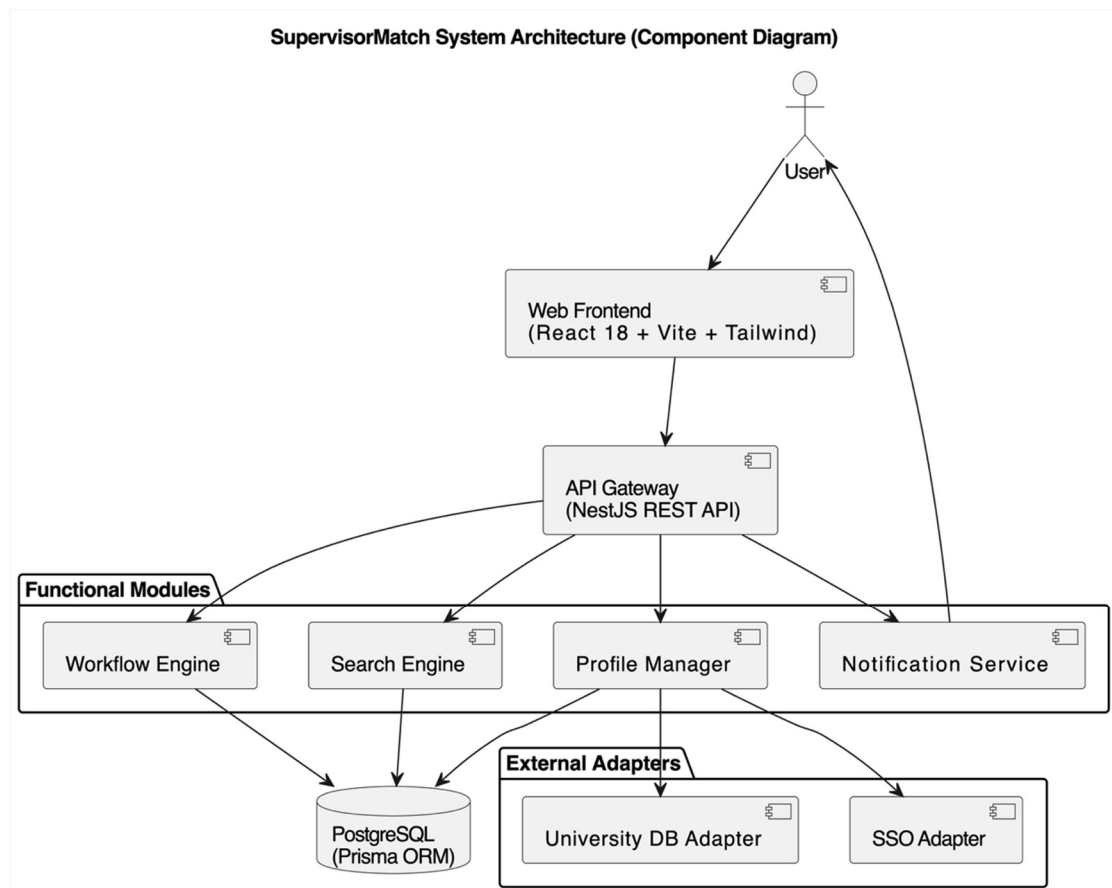
5.1 Development Approach

The team follows an Agile development methodology, organized into two-week sprints. Each sprint includes a planning session, daily stand-ups, a sprint review, and a retrospective. Sprint artefacts (backlog, burn-down chart) are maintained in the team's project board.

5.2 System Architecture

SupervisorMatch follows a Modular Monolith architectural style. This choice prioritises maintainability and lower deployment complexity over the distribution overhead of a microservices approach — appropriate for the current team size and academic timeline. The architecture consists of the following layers:

- Web Frontend — React 18 with Vite bundler and Tailwind CSS. Responsible for all user-facing interactions.
- API Gateway — NestJS application exposing a RESTful API. Routes requests to functional modules.
- Functional Modules — Search Engine, Workflow Engine, Profile Manager, Notification Service. Each module is independently testable.
- External Adapters — Isolated integration points for University SSO and the University Database, ensuring the core business logic remains decoupled.
- Database Layer — PostgreSQL managed via Prisma ORM. Full-text search indices are defined on supervisor name, department, and research area fields.



5.3 Technology Stack

Layer	Technology	Justification
Frontend	React 18 + Vite + Tailwind CSS	Industry-standard SPA framework; Vite provides fast HMR; Tailwind enables rapid UI development.
Backend	NestJS (Node.js)	TypeScript-native framework with strong modularity; built-in support for guards, interceptors, and dependency injection.
Database	PostgreSQL + Prisma ORM	Relational model ensures strong consistency for availability data (ADR-01); Prisma provides type-safe queries.
Authentication	JWT + Role Guards	Stateless, scalable authentication; role guards enforce RBAC at controller level.
Search	PostgreSQL Full-Text Search	Avoids additional infrastructure (e.g., Elasticsearch) while meeting sub-second latency requirement at expected scale.
Deployment	Vercel (Frontend) + Railway/Render (Backend) + Supabase (DB)	Low-cost, zero-DevOps hosting suitable for academic project timeline.
Design	Figma	Collaborative wireframing and prototype validation with target users.
Documentation	Microsoft Word + PowerPoint	Institutional requirement for pre-defence submission.

5.4 Architectural Decision Records (ADRs)

ADR-01: Relational Database over NoSQL

Context: The team considered MongoDB for its flexible schema and ease of setup. However, supervisor availability (a numeric slot count) requires strong ACID consistency — a concurrent accept operation from two students must not result in over-booking.

Decision: PostgreSQL with row-level locking on capacity updates.

Consequence: Slightly more rigid schema migration process; mitigated by Prisma migrations.

ADR-02: Secure Contact Exchange

Context: Revealing supervisor email addresses on public search results would expose faculty to unsolicited contact and undermine the request workflow.

Decision: Contact details are withheld at the API level (not merely hidden in the UI) until a request reaches ACCEPTED status.

Consequence: Requires an additional API endpoint for post-acceptance profile detail; adds a small complexity overhead.

ADR-03: Modular Monolith over Microservices

Context: Microservices would add network overhead, distributed tracing complexity, and deployment pipeline requirements that exceed the team's current capacity.

Decision: Single deployable NestJS application with internally separated modules.

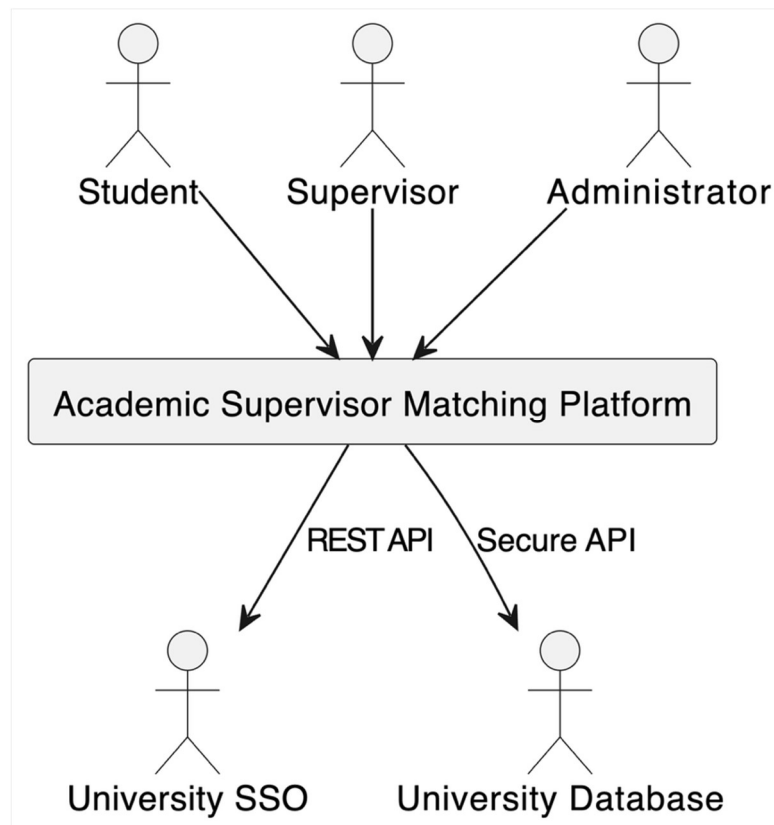
Consequence: Easier local development and testing; future migration to microservices remains possible by extracting modules along existing boundaries.

6. MVP and UML Diagrams / Architecture

This section documents the platform's architecture and key workflows through UML diagrams. Insert your exported diagrams in the marked locations below.

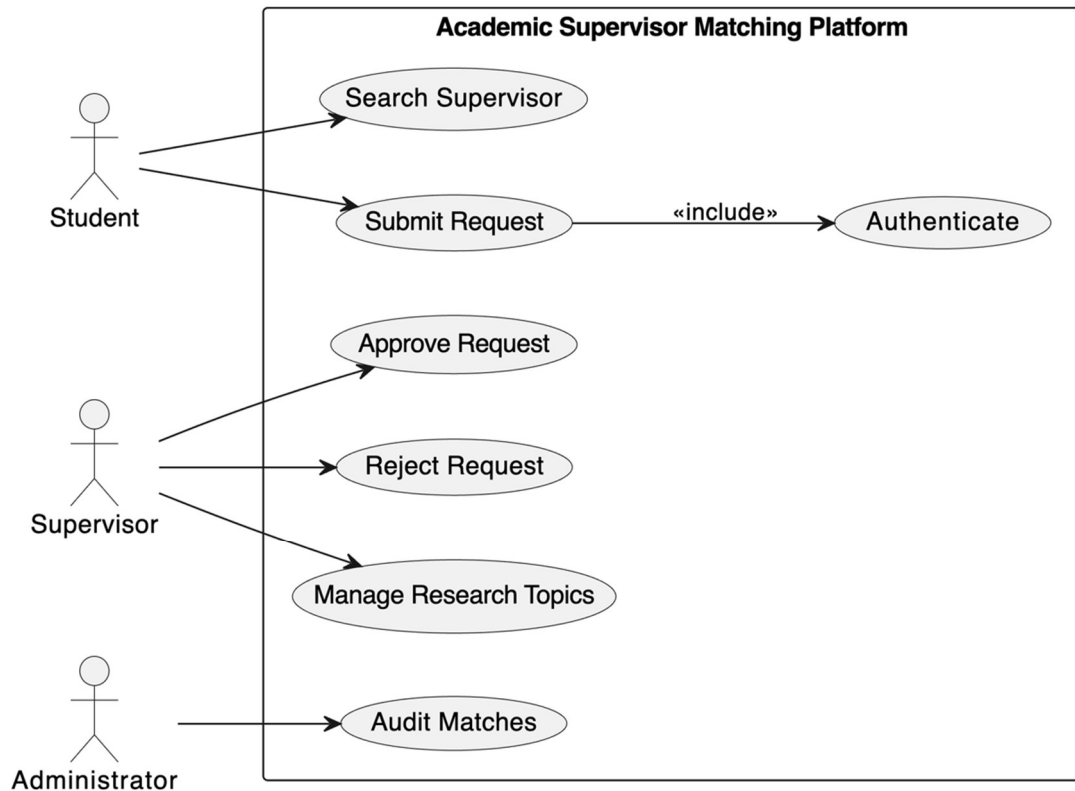
6.1 Context Diagram

The context diagram illustrates the system boundary and its interactions with external actors: Student, Supervisor, Administrator, University SSO, and University Database.



6.2 Use Case Diagram

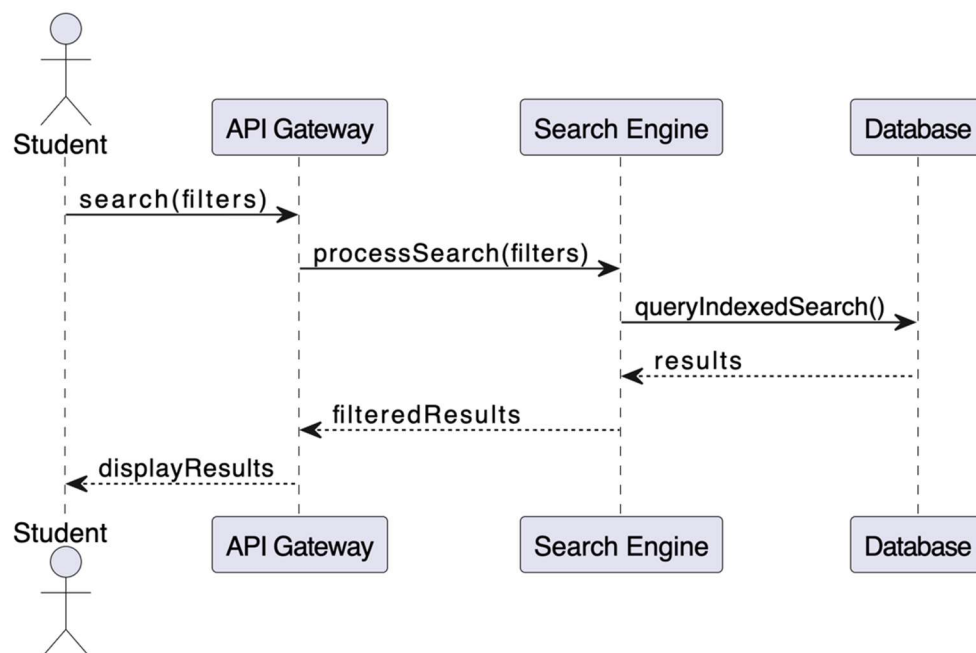
The use case diagram captures the functional interactions of each actor with the system. Key use cases include: Search Supervisor, Submit Request (includes Authenticate), Approve/Reject Request, Manage Research Topics, and Audit Matches.



6.3 Sequence Diagrams

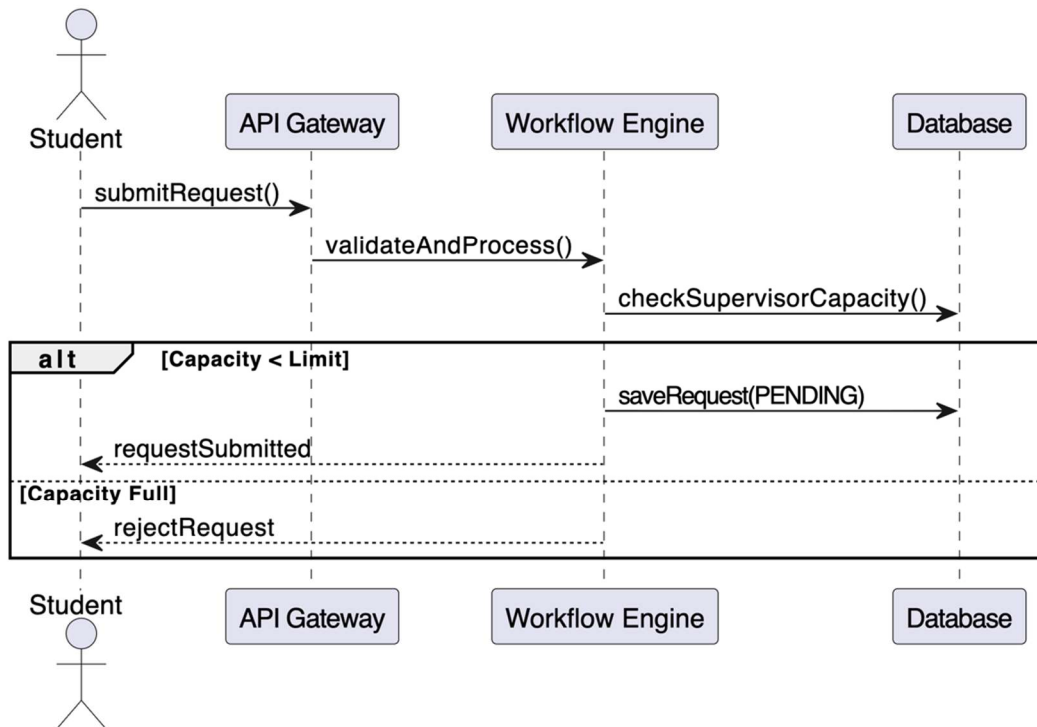
6.3.1 Search Workflow

The search sequence shows a Student submitting filters to the API Gateway, which delegates to the Search Engine; the Search Engine queries the indexed database and returns filtered results.



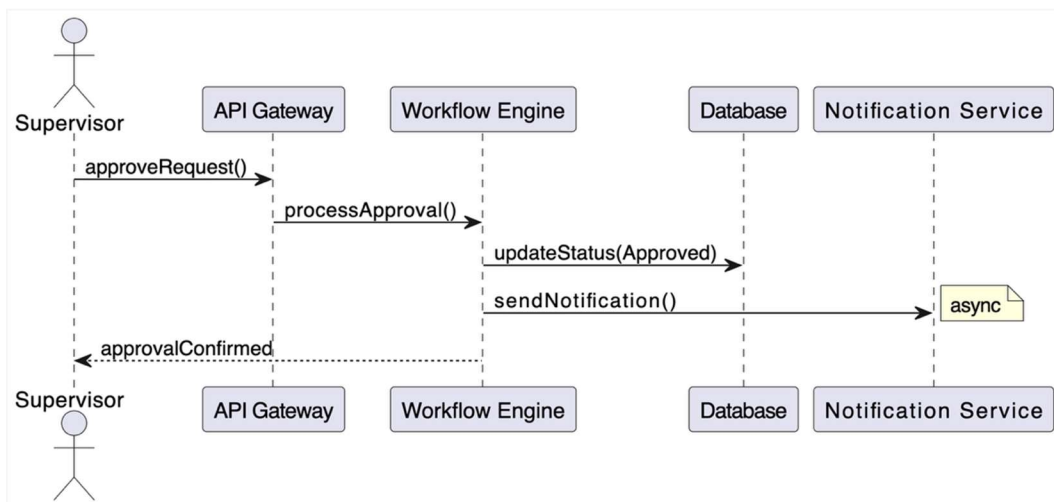
6.3.2 Submit Request Workflow

This diagram demonstrates the Alt-block capacity check: if supervisor capacity is below the limit, the request is persisted with status PENDING; if capacity is full, the request is rejected immediately.



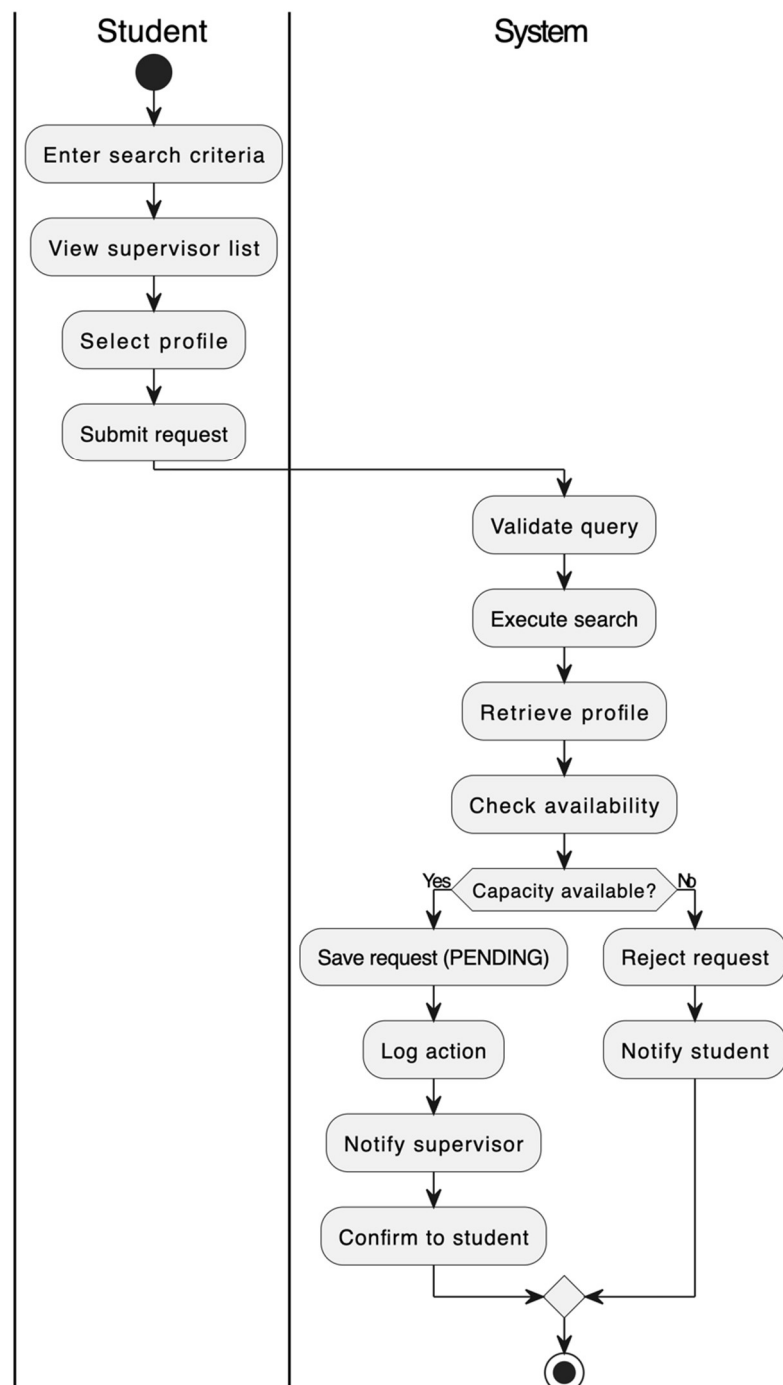
6.3.3 Approval Workflow

The approval sequence shows the supervisor approving a request, the Workflow Engine updating the status and asynchronously triggering a notification (keeping the UI responsive).

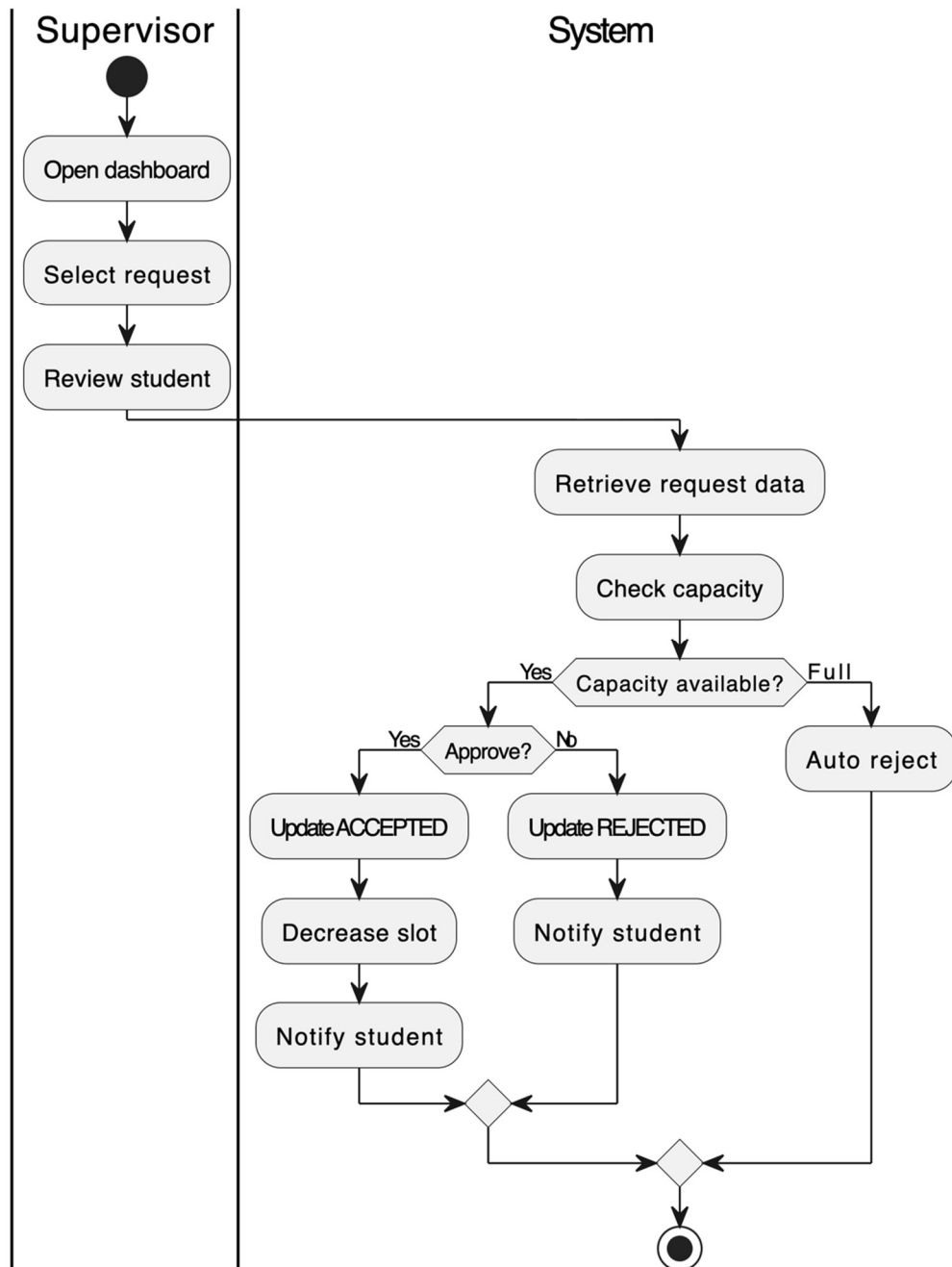


6.4 Activity Diagrams

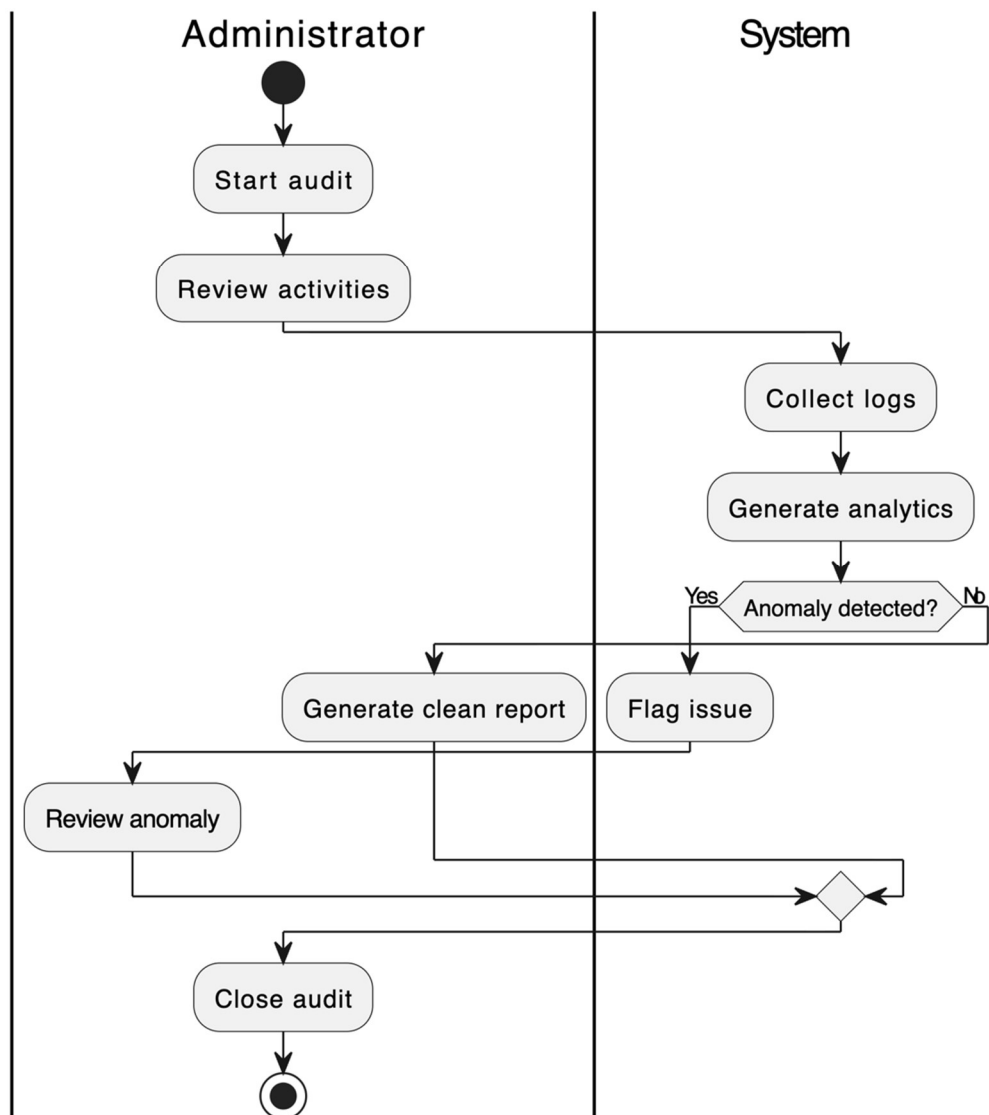
6.4.1 Student Supervisor Search and Request Submission



6.4.2 Supervisor Request Evaluation



6.4.3 Administrator Audit Workflow



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6.5 MVP Mockups

The following mockups represent the core student and supervisor interfaces validated through a Figma prototype and informal usability testing with AITU students.



Welcome Back

Sign in to SupervisorMatch

Email

Password

Login

Don't have an account? [Register here](#)

Demo Accounts

Student: **student@test.com** (or any new student email)

Supervisor: **johnson@university.edu**

Password: any password

SupervisorMatch
Student

AboutProfileLogout

SearchRequests

Hi, Alex Carter
Find your perfect supervisor with smart filters and real-time availability.

Search by name, field, department...

All DepartmentsAll Areas

☐ Available only

6 supervisors



Dr. Sarah Johnson

Professor
Computer Science

Specializing in AI and ML with 15+ years of research experience.

Availability

3/8

Artificial Intelligence

Machine Learning

Deep Learning

Write a short message to the



Prof. Michael Chen

Associate Professor
Data Science

Works on scalable analytics pipelines and real-time data intelligence systems.

Availability

1/8

Data Mining

Big Data Analytics

Knowledge Graphs

Write a short message to the



Dr. Emma Williams

Senior Lecturer
Software Engineering

Explores resilient cloud platforms, observability, and CI/CD automation.

Availability

5/8

Cloud Computing

DevOps

Distributed Systems

Write a short message to the



Dr. Omar Hassan

Professor
Electrical Engineering

Leads projects in secure embedded design and real-time smart infrastructure.

Availability

2/8

Embedded Systems

IoT Security

Cyber-Physical Systems

Write a short message to the

SupervisorMatch
Student

AboutProfileLogout

SearchRequests

My Requests

Track your supervisor requests across pending, under review, accepted, and rejected states.

Dr. Sarah Johnson

Computer Science

Under Review

I am interested in AI for education and would like to work with your group.

Sent on 3/17/2026, 4:10:00 PM

Remove

Dr. Emma Williams

Software Engineering

Accepted

My thesis idea is cloud-native tooling for student collaboration systems.

Sent on 3/10/2026, 2:45:00 PM

Contact available: Dr. Emma Williams | Software Engineering

Remove

Prof. Michael Chen

Data Science

Rejected

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My Profile

Student



Alex Carter
student@test.com
Student

Full Name

Alex Carter

Email (Login)

student@test.com

Phone Number

+10000000000

Study Level

Final Year

Department

Computer Science

Group Name

CS-2024-A

Research Interests

Artificial Intelligence, Learning Analytics

Bio

Student interested in AI-supported education and recommendation systems.

Dashboard



Dr. Sarah Johnson
Professor - Computer Science
Artificial Intelligence Machine Learning Deep Learning

Slots

3/8

Pending

0

Topics

2

Requests

Topics

balzhankomekbaeva
balzhankomekbaeva@gmail.com
I would like to discuss potential supervision opportunities.
Received on 3/26/2026, 10:45:26 PM

Under Review Accept Reject

Alex Carter
student@test.com
I am interested in AI for education and would like to work with your group.
Received on 3/17/2026, 4:10:00 PM

Under Review

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Dashboard

Dr. Sarah Johnson
Professor · Computer Science

Artificial Intelligence Machine Learning Deep Learning

Slots **3/8** Pending **0** Topics **2**

Requests **Topics**

Research Topics
Manage your available thesis and diploma topics.

Topic Title
e.g., Neural Networks for Image Recognition

Main Area
e.g., Machine Learning

Description
Briefly describe this topic for students

+ Add Topic

7. Technology Comparison

This section provides a systematic comparison of the technologies evaluated or considered for each major layer of the system, justifying the final selections.

7.1 Frontend Framework

Framework	Pros	Cons	Selected?
React 18	Largest ecosystem; team familiarity; strong TypeScript support; Vite integration	Requires additional libraries for routing and state (React Router, Zustand)	YES
Vue 3	Gentler learning curve; built-in reactivity	Smaller ecosystem; less team familiarity	No
Angular	Opinionated structure; built-in DI	Steeper learning curve; heavier bundle	No

7.2 Backend Framework

Framework	Pros	Cons	Selected?
NestJS	TypeScript-native; modular architecture aligns with project structure; built-in guards and pipes	More boilerplate than Express	YES

Express.js	Minimal, fast; large community	No structure enforced; requires manual wiring	No
Django (Python)	Batteries-included; ORM built in	Team lacks Python proficiency for backend	No

7.3 Database

Database	Pros	Cons	Selected?
PostgreSQL	ACID compliance; full-text search; strong consistency for capacity counters	More complex setup than MongoDB	YES
MongoDB	Flexible schema; easy initial setup	Weaker consistency guarantees; less suitable for relational data	No
MySQL	Mature; widely supported	No built-in full-text search as powerful as PostgreSQL; less Prisma optimization	No

8. Data Collection

Primary Research

In order to validate the problem statement and refine requirements, the team conducted primary data collection during February–March 2026. The following method were used:

Survey

A structured survey was distributed to 40 AITU students via the university's internal communication channel. Results (n = 34 respondents):

Question	Majority Response	Supporting Insight
How easy was it to find supervisors' research interests?	Moderate difficulty	Most responses around 2–3 (partially available information)
Was supervisor workload information available?	Partially / Not available	Majority reported limited or no visibility
How many supervisors were contacted before approval?	2 or more	Many students contacted multiple supervisors
What difficulties were experienced?	Multiple issues reported	Most common: overload, slow response, lack of transparency
How long did the process take?	1–2 weeks or more	Several cases exceeded 2 weeks
How transparent is the process?	Moderately low	Most ratings between 2–3 out of 5

9. Testing Strategy

Testing is planned across four levels, following the Agile testing quadrants approach. The table below maps each test type to its scope, tools, and target NFR.

Test Type	Scope	Tool	Target Requirement
Unit Testing	Individual NestJS services and controllers	Jest	NFR-MAINT-01 ($\geq 70\%$ coverage)
Integration Testing	Frontend–Backend API interaction; database transactions	Supertest + Jest	NFR-SEC-01, NFR-SEC-02, NFR-SEC-03
Performance Testing	Search endpoint under concurrent load	k6 / Artillery	NFR-PERF-01 ($< 1\text{s}$ latency), NFR-PERF-02 (200 concurrent)
Usability Testing	End-to-end student and supervisor workflows	Moderated sessions with 5 participants	NFR-USAB-01 (task completion $< 3\text{ min}$)

10. Project Plan and Current Progress

10.1 Overall Timeline

Stage	Key Tasks	Timeline	Status
Problem Analysis & Requirements	Stakeholder interviews, survey, requirements specification	February 2026	Complete
Design Phase	Figma wireframes, system architecture design, team roles	March 2026	Complete
Backend Development	NestJS setup, API development, PostgreSQL + Prisma integration, JWT auth, feature implementation	April 2026	In Progress
Frontend Development	React + Vite setup, UI implementation, API integration, responsive design	April 2026	In Progress
Testing & Deployment	Unit, integration, performance, usability testing; cloud deployment	May 2026	Planned
Final Preparation	Documentation, second pre-defence report, final defence presentation	May 2026	Planned

10.2 Individual Contributions

Team Member	Role	Responsibilities
Balzhan Komekbaeva	Analytics Lead	Requirements analysis; system logic design; team coordination; project documentation; defence preparation
Damir Bekturov	Backend Developer	NestJS architecture; API development (auth, search, requests); PostgreSQL schema with Prisma; JWT and RBAC; backend testing
Tairlan Zhanashev	Frontend Developer	React + Tailwind UI implementation; UX navigation and layouts; API integration; responsive design; frontend testing

11. Conclusion

This report has presented the expanded technical and analytical foundation for the SupervisorMatch platform, addressing all points of feedback raised at the First Pre-Defence:

1. The institutional context has been clearly defined: the platform targets Kazakhstani universities, initially AITU, serving Student, Supervisor, and Administrator roles whose pain points have been documented through primary data collection.
2. A comprehensive requirements specification has been provided, distinguishing 22 functional requirements across seven feature areas and 9 non-functional requirements across performance, security, usability, maintainability, and scalability.
3. A structured analysis of four existing platforms (FindASupervisor, ARAP, SupervisorFinder, and informal LinkedIn/email processes) demonstrates the gap that SupervisorMatch fills — specifically the absence of a governed, end-to-end request workflow with real-time availability and administrative oversight.

The team is currently in the active development phase (April 2026), with backend API endpoints for authentication, search, and request management under implementation, and the React frontend being developed in parallel. The platform is on track for deployment and usability testing in May 2026, ahead of the Final Defence.

Diagrams produced as part of the Architecture course (Component, Use Case, Sequence, and Activity diagrams) are to be inserted in the marked locations in Section 6. The team is prepared to submit these alongside the completed document upon instructor confirmation.

12. References

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